

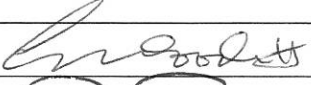
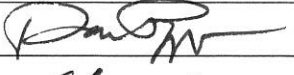
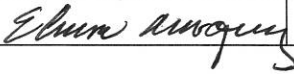
		CONTROLLED FORM		
TITLE: GXP and Non-GXP Study Report Review and Approval Form				
DOCUMENT NO. GE003FM01	VERSION 02	ISSUE DATE 17 Oct 2014	EFFECTIVE DATE 20 Oct 2014	PAGE 2 of 2

☐ Vendor Study Protocol ☒ Vendor Study Report ☐ Lpath Study Report					
Lpath Study No.: LT3114-PHA-019-R		Version No.: 01		☒ Non GXP ☐ GCP	☐ GLP ☐ GMP
Product Name: LT3114		Test Article: LT3114; LT1023			
Study Title: Lpathomab: Evaluation of efficacy of intravenous administrations in a rat model of S-antigen-induced experimental autoimmune uveitis					
Study Vendor/Site: ☐ N/A Iris Pharma	Study Director: ☐ N/A Solveig Lefranc-Jullien	Study Contributor(s): ☒ N/A			
Vendor Study No./Version: ☐ N/A L15U27313		Study Dates: February 2014		Report Date: October 2014	
Lab Notebook Reference: LNB#278 pp 40-42; #308 pp 142-149; #311 pp 33-35; #328 pp105-109 ☐ N/A					

Executive summary of study results: ☒ See Attached ☐ N/A

The efficacy of LT3114 was examined in a model of uveitis in female Lewis rats. Ocular inflammation was induced by injection in the footpad of 100ul of human S-Ag (100ug) in Freund's complete adjuvant (2mg/ml) and an i.p. injection of 1ug/100ul pertussis toxin. LT3114 (15 and 30 mg/kg) and LT1023 (30 mg/kg) were administered i.v. at 7 days post induction continuing twice weekly until the end of the study (15 animals/group). A positive control, cyclosporin A (25mg/kg) was given orally once daily starting at 7 days post induction.

High mortality was observed between day 14 and 17 in all groups treated with Ab (LT3114 - 27% and 47% mortality for the low and high doses respectively or control antibody - 67% mortality) resulting in early termination of the study. The mortality in all Ab groups suggested that the effects seen were not LT3114 specific but a limited investigation into the activity and PK of these batches of Ab was carried out and is noted in the attached Appendix. Based upon this data LT3114 in the batches sent to Iris and returned showed no decrease in activity and no increase in aggregates/particulates. The PK data from the samples collected at Iris on termination of the available animals showed that there was no obvious increase in clearance of the drug. Limited data on anti-drug Ab estimation showed some elevation of anti-drug antibodies particularly in the LT1023 group. However, it must be noted that the animals were specifically primed to produce inflammation and this coupled with dosing of Ab may have led to the issues encountered.

LPATH APPROVAL			
Function	Name (Print)	Signature	Date
Study Director/Monitor	N/A		
Contributor	N/A		
Research	GARY WOODWORTH		23 rd December 2014
Development	DARIO A. PAGGIARINO		23 DEC, 2014
Quality Assurance	ELVIRA MARQUEZ		23 DEC 14

STUDY REPORT: LT3114-PHA-019-R

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Appendix

In follow up to a study conducted at Iris (Study number: L15U27313), LPath carried out a number of investigations using LT3114 and the control antibody, LT1023. The batches of material that were used in the LPath investigations were the same as those sent to Iris but were stored at LPath as well as material that was returned from Iris (LT3114 batches LP121416 and LP121417 and control antibody – LT1023 batches LP101210 and LP121424).

For examination of the physical integrity of the dosing solutions, samples of the dosing solutions returned from Iris were tested using the following:

- Ligand binding (ELISA)
- SEC (examination of aggregate)
- Visual inspection (black background) for particulates
- SDS-PAGE (reduced and non-reduced)
- Endotoxin

In all of these tests, test results from both LT3114 batches returned from Iris were within the original specifications indicating no alteration in protein during transfer (and return).

Batches (LT3114 and control antibody) that were stored at Lpath were administered to three Sprague Dawley rats per group at the highest dose administered by Iris (30 mg/kg) three times in one week (identical to the Iris dosing regimen) and no adverse effects were observed in any of the LT3114 or control antibody animals.

Upon receiving the batches returned from Iris, two Sprague Dawley rats per group received the same dosing regimen as that used in the Iris studies of control antibody (30mg/kg, three times a week, Batch LP121424) and LT3114 (25mg/kg, three times a week, Batch LP121417). No adverse effects were seen following any of the administered doses. The plasma pharmacokinetics of LT3114 from the returned batch of LT3114 was similar to that reported previously (Report number: LT3114-PHA-010-R).

Plasma samples obtained from animals dosed at Iris were analyzed for human IgG content and results are tabulated below (Table 2). Concentrations of LT3114 in rats sacrificed on day 18 ranged between 40 µg/ml and 247 µg/ml in the high dose group and between 1 µg/ml and 170 µg/ml in the low dose group. Concentrations of LT1023 ranged between 50 µg/ml and 280 µg/ml in the isotype control group. Samples taken prior to 18 days tended to have the higher concentrations of the range (220-270 µg/ml for the high dose LT3114; 22 µg/ml for the low dose LT3114 and 52-321 µg/ml for the isotype control) which would be expected as

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they were sampled at various times before day 18 and under less than ideal conditions. Previous single dose data showed that at 72h post dose LT3114 plasma concentrations were approximately 50-60 µg/ml at 10 mg/kg and declined to 30-50 µg/ml at 96h. Based on this limited data, plasma concentrations of antibody were similar to those observed previously except for a few outliers. Thus, there is no evidence of increased clearance of LT3114 or excessive accumulation of LT3114 in plasma when administered three times a week in Sprague Dawley rats.

The presence of anti-ADA antibodies was detected by a direct-binding immunoassay (ELISA) method. LT3114 and LT1023 antibodies were diluted to 1 µg/mL in carbonate buffer (100 mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). Plates (384 well) were coated with 15 µL/well of coating antibody solution and incubated at 4°C overnight. The next morning, plates were emptied and 25 µL/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) was added and incubated for 1 hour at room temperature. Plates were washed 4 times with PBS, and 15 µL of diluted test samples per well was added in duplicate and incubate for 1 hour at room temperature. Rat plasma samples were diluted with blocking buffer to 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490. Then, plates were washed 4 times with PBS and 15 µL of 1:10,000 blocking buffer diluted goat anti-rat IgG HRP (Jackson 112-035-062) was added to each well and incubate for 1 hour at room temperature. Finally, the plates were washed 4 times with PBS and developed with 15 µl/well TMB substrate at 4°C. After 5 minutes, the reaction was stopped with 1M H₂SO₄, 15 µl/well and the OD was measured at 450 nm. Data was analyzed using Graphpad Prism software. Not all animals had sufficient sample to test for anti-drug antibodies. Data from study L15U27313, using Lewis albino rats, suggest that in all antibody treated animals (LT3114 and LT1023), there were increased anti-drug antibody titers although these were variable and did not necessarily correlate with antibody dosage (Figure 1). In general, antibody titers were higher in the control antibody animals (LT1023) when compared with LT3114.

Table 1 Animal numbers associated with study L15U27313*

Treatment	Dose	Animal IDs
LT3114	30 mg/kg	1-15
LT3114	10 mg/kg	16-30
LT1023	30 mg/kg	31-45
Cyclosporin A	25 mg/kg	46-60

* Not all animals in these groups had plasma/serum assayed for drug concentration or anti-drug antibody titer

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**Table 2 Concentrations of LT3114 and LT1023 at termination of study
L15U27313**

Treatment	Dose	Animal number	Observations	[mAb] ug/mL
LT3114	30 mg/kg	1	Sacrificed on Day 18	58
		2	Sacrificed on Day 18	247
		3	Found dead on Day 15	
		4	Found moribund on Day 16 and then sacrificed	
		5	Found dead on Day 17	215
		6	Found moribund on Day 15 and then sacrificed	268
		7	Found dead on Day 15	
		8	Sacrificed on Day 18	79
		9	Sacrificed on Day 18	43
		10	Sacrificed on Day 18	170
		11	Sacrificed on Day 18	49
		12	Found dead on Day 16	
		13	Sacrificed on Day 18	169
		14	Sacrificed on Day 18	93
		15	Found moribund on Day 16 and then sacrificed	215
LT3114	15 mg/kg	16	Sacrificed on Day 18	65
		17	Sacrificed on Day 18	88
		18	Sacrificed on Day 18	1
		19	Sacrificed on Day 18	126
		20	Found on Day 16	
		21	Sacrificed on Day 18	20
		22	Sacrificed on Day 18	167
		23	Sacrificed on Day 18	38
		24	Found dead on Day 16	
		25	Sacrificed on Day 18	114
		26	Sacrificed on Day 18	96

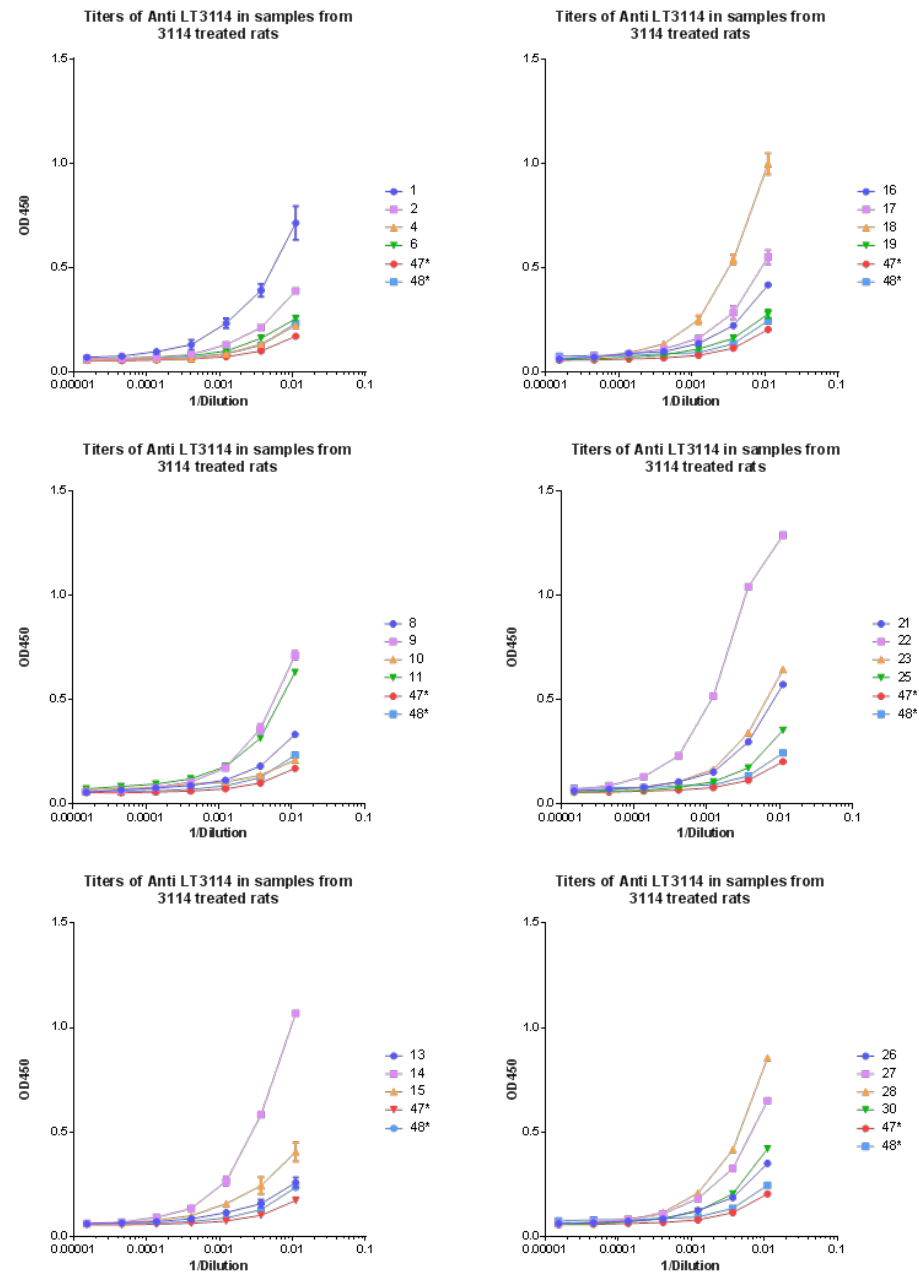
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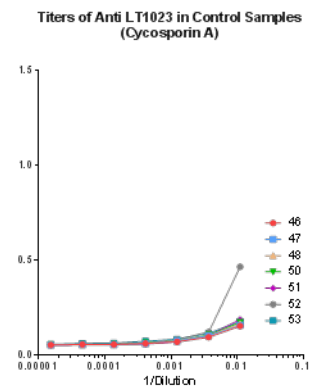
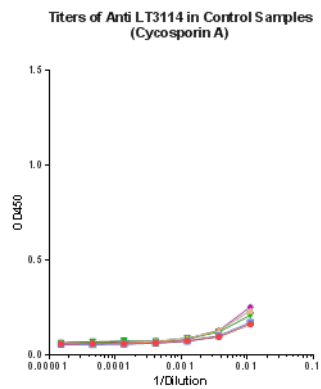
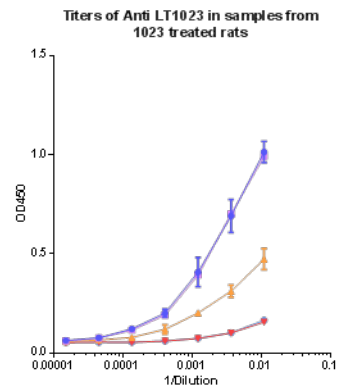
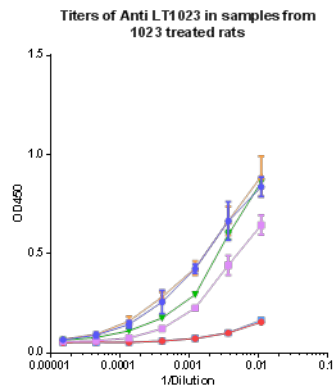
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**Table 2 Concentrations of LT3114 and LT1023 at termination of study
L15U27313**

Treatment	Dose	Animal number	Observations	[mAb] ug/mL
		27	Found moribund on Day 17 and then sacrificed	22
		28	Sacrificed on Day 18	10
		29	Found dead on Day 16	
		30	Sacrificed on Day 18	76
LT 1023	30mg/kg	36	Found moribund on Day 15 and then sacrificed	207
		37	Sacrificed on Day 18	167
		38	Found dead on Day 14	
		39	Found dead on Day 16	
		40	Sacrificed on Day 18	51
		41	Found dead on Day 16	52
		42	Sacrificed on Day 18	279
		43	Found moribund on Day 15 and then sacrificed	321
		44	Sacrificed on Day 18	279
		45	Found moribund on Day 15 and then sacrificed	321

Figure 1 Anti-LT3114 and Anti-LT1023 antibody titers in animal serum from study L15U27313.

* Controls from Cyclosporin A group



LT3114 and LT1023 antibodies were diluted to 1 µg/mL to coat plates and the ELISA was carried out as described above. Rat plasma samples from LT3114, LT1023, and cyclosporin A treated groups from study L15U27313 were diluted with blocking buffer to 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1: 7290, 1: 21870, 1: 65610, 1: 196380 and 1: 590490. Control Samples 47 and 48 were run in all plates as negative controls. Data was analyzed using Graphpad Prism software.

STUDY REPORT: LT3114-PHA-019-R	
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Title: Lpathomab: Evaluation of efficacy of intravenous administrations in a rat model of S-antigen-induced experimental autoimmune uveitis	
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DATA STORAGE

Testing of 3114 and 1023 in Sprague Dawley rats: LPath Notebook No.278, Page40-42

Anti-human IgG ELISA: LPath Notebook No.308, Page142-149

Direct ELISA, SDS-PAGE, Endotoxin: LPath Notebook No. 311, Page 33-35

SE-HPLC: LPath Notebook No. 328, Page 105-109



LPATHOMAB™

***EVALUATION OF EFFICACY OF INTRAVENOUS ADMINISTRATIONS IN A RAT
MODEL OF S-ANTIGEN-INDUCED EXPERIMENTAL AUTOIMMUNE UVEITIS***

Study Number: L15U27313

Non-GLP Study

Sponsor Representative for ***LPATH, INC.***: Dario PAGGIARINO, M.D.

Study Director for ***IRIS PHARMA***: Solveig LEFRANC-JULLIEN, Ph.D.

Report version 4 (Final version) - 27 pages

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1. SUMMARY

Purpose:

The aim of this study was to evaluate the effect of Lpathomab™ at two dose levels after twice weekly intravenous administrations in a rat model of EAU.

Method:

Sixty (60) Lewis albino rats were divided in 4 groups of 15 rats (10 main animals and 5 satellites). EAU was induced in all animals by injection of S-Antigen in Complete Freund's adjuvant on Day 0.

The schedule is presented in **Table 1** hereafter.

Table 1: Study design

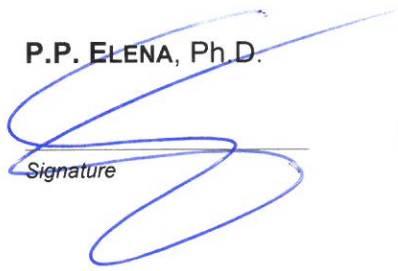
Study date	Procedure	Ocular examination	Sampling
Baseline	General clinical examination - Weighing	Slit-lamp	-
D0	General clinical examination Induction of ocular inflammation	-	-
D1 to D6	General clinical examination	-	-
D7	General clinical examination - Weighing Intravenous administration of test and control items Once daily oral administration of reference item	-	-
D8 to D9	General clinical examination Once daily oral administration of reference item	-	-
D10	General clinical examination Intravenous administration of test and control items Once daily oral administration of reference item	-	-
D11	General clinical examination Once daily oral administration of reference item	-	-
D12 to D13	General clinical examination Once daily oral administration of reference item	Slit-lamp for main animals	-
D14	General clinical examination - Weighing Intravenous administration of test and control items Once daily oral administration of reference item		-
D15	General clinical examination Once daily oral administration of reference item		-
D16	General clinical examination		-
D17	General clinical examination	-	-
D18	General clinical examination – Weighing Euthanasia	Slit-lamp for all animals (main + satellites)	• Whole blood (for plasma) • Vitreous (V), choroid/retina (CHR) and iris-ciliary body (ICB)

Results/ Conclusion:

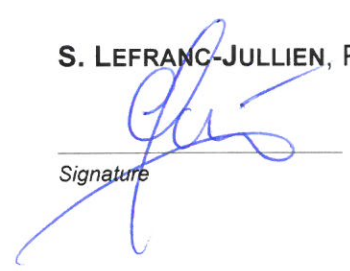
A high mortality was observed between Day 14 and Day 17 in groups treated with Lpathomab™ isotype control at 30 mg/kg (10 animals found dead or moribund out of 15) and Lpathomab™ at 30 mg/kg (7 animals found dead or moribund out of 15). For the group treated with Lpathomab™ at 15 mg/kg, the mortality was moderate (4 animals found dead or moribund out of 15). No reason was found to explain the observed mortality. With the sponsor agreement, the administrations of all study treatments (test, control and reference) and the ocular examinations were discontinued earlier than originally planned.

As a result of the high mortality rate and early discontinuation of study treatments, no conclusions could be drawn regarding the potential effect of Lpathomab™ on intraocular inflammation regardless of the dose.

No abnormal behaviour or other abnormal signs were observed in the treatment arm receiving Cyclosporin A during the study period.

2. STATEMENTS**LPATHOMAB™****EVALUATION OF EFFICACY OF INTRAVENOUS ADMINISTRATIONS IN A RAT
MODEL OF S-ANTIGEN-INDUCED EXPERIMENTAL AUTOIMMUNE UVEITIS****MANAGEMENT AND STUDY DIRECTION****Director of Test Facility:** P.P. ELENA, Ph.D.
SignatureOctober 13th, 2014
Date

I, the undersigned, declare that this study was performed according to the study plan and I assume the responsibility of the validity of reported data.

Study Director: S. LEFRANC-JULLIEN, Ph.D.
SignatureOctober 13th, 2014
Date

3. GENERAL INFORMATION

3.1. TEST FACILITY

Study Direction**Address**

IRIS PHARMA
Les Nertières
Allée Hector Pintus
FR-06610 LA GAUDE

Study Director

Solveig LEFRANC-JULLIEN, Ph.D.
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Tel: +33 493 59-4959
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e-mail: s.lefranc-jullien@iris-pharma.com

3.2. SPONSOR

Address

LPATH, INC.
4025 Sorrento Valley Blvd.
San Diego, CA 92121
USA

Sponsor Representative

Dario PAGGIARINO, M.D.
LPATH, INC.
Tel: 858 678 0800
Fax: 858 678 0900
e-mail: DPaggiarino@lpath.com

3.3. GENERAL INFORMATION

3.3.1. Study Dates

Study plan version 2 (final version) signed on:	February 07, 2014
Start of In-life period:	February 06, 2014
End of In-life period:	February 27, 2014
Report version 1:	June 20, 2014
Report version 2:	August 12, 2014
Report version 3:	September 29, 2014
Report version 4:	October 13, 2014

3.3.2. Contractual Clauses

All employees of Iris Pharma are contractually bound by a confidentiality clause.

4. INTRODUCTION

Uveitis, the inflammation of the uveal tract, can be classified anatomically as anterior, intermediate, posterior, or diffuse (panuveitis) depending on the segment of the eye that is affected. Each year, approximately 17% of active uveitis patients experience some degree of vision loss. Uveitis may be idiopathic, associated with systemic diseases such as Behçet's disease and Vogt-Koyanagi-Harada disease, or resulting from a variety of infectious agents. Inflammation resulting from uveitis may lead to conditions like cataract, glaucoma, and cystoid macular oedema that may cause irreversible vision loss. Although the exact cause of the disease is still unknown, a wide array of studies using different animal models have established that the inflammation in uveitis is due to an autoimmune response to various ocular antigens. Most studies in the past three decades have focused on the role of T cells in the pathogenesis of uveitis [1, 2, and 3]. Specific antigenic preparations, originating from the tissue where the pathology is directed to, have been used to model the different types of human uveitis (e.g. anterior versus posterior). Accordingly, EAU (Experimental Autoimmune Uveitis) is typically induced by immunizing susceptible animals with one of the several Ag originating from the retina. The EAU model induced with retinal arrestin (S-Ag) has been the most widely used and the best characterized rat model. The onset of disease is observed between Days 9 and 12 after immunization. The clinical course of the disease is typically monophasic. Clinical signs of the disease are assessed by fundus examination and scored using a semi-quantitative scale. Uveoretinal inflammation and retinal damage are graded on histological preparations.

The EAU model has been instrumental in investigating the biochemical and cellular mechanisms implicated in uveitis as well as the pharmacology of potential drug

candidates for the treatment of uveitis and ocular inflammation. For instance, previous studies have demonstrated that development of EAU in rats is prevented by the immunosuppressant cyclosporine A [4], a selective modulator of T-lymphocyte functions.

5. OBJECTIVE

The aim of this study was to evaluate the effect of Lpathomab™ at two dose levels after twice weekly intravenous administrations in a rat model of EAU.

6. MATERIAL

Table 2: Material table – Test item

Material	Test item
Denomination	Lpathomab™ / LT3114
Concentration	Lot LP121416, 20.1 mg/mL; Lot LP121417, 21.7 mg/mL
Characteristics (form, aspect, molecular weight, etc.)	Frozen liquid, 147 kDa
Supplier	LPATH, INC.
Reference and/or batch numbers	LP121416 and LP121417
Purity	> 90%
Expiry date	08/01/2014 (6 months stored at 2-8°C)
Quantity received	750 mg
Preparation (concentration, vehicle, composition, appearance after formulation, pH)	15 and 30 mg/kg
Storage conditions and stability (before/after preparation, before/after opening)	2-8°C for all
Safety (powder, formulation)	Formulated in phosphate buffered saline
Documentation	No certificate of analysis and stability was provided by the Sponsor

Table 3: Material table – Control item

Material	Control item
Denomination	Lpathomab™ isotype
Concentration	Lot LP101210, 15.3 mg/mL; Lot LP121424 19.5 mg/mL
Characteristics (form, aspect, molecular weight, etc.)	Frozen liquid, 150 kDa
Supplier	LPATH, INC.
Reference and/or batch numbers	Lot LP101210 and Lot LP121424
Purity	< 90%
Expiry date	08/01/2014 (6 months stored at 2-8°C)
Quantity received	500 mg
Preparation (concentration, vehicle, composition, appearance after formulation, pH)	30 mg/kg
Storage conditions and stability (before/after preparation, before/after opening)	2-8°C for all
Safety (powder, formulation)	Formulated in phosphate buffered saline
Documentation	No certificate of analysis and stability was provided by the Sponsor

Table 4: Material table – Reference item

Material	Reference item
Denomination	Sandimmune® (Cyclosporin A)
Supplier	Novartis
Batch number	H5009B
Preparation (concentration, vehicle, composition, appearance after formulation, pH)	25 mg/kg/day – 1 mL/kg One preparation every two days, 1 mL in 3 mL olive oil
Storage conditions	Room temperature
Documentation	CA, MSDS, CS formulation protocol in raw data

Disposition of test item:

Following the completion of the In-life period, empty containers of test item were discarded according to Iris Pharma standard operating procedures. The remainder of test item was sent back to the Sponsor.

Disposition of control item:

Following the completion of the In-life period, empty containers of control item were discarded according to Iris Pharma standard operating procedures. The remainder of control item was sent back to the Sponsor.

Disposition of reference item:

Following the completion of the In-life period, empty containers of reference item were discarded according to Iris Pharma standard operating procedures.

7. ANIMALS AND ANIMAL HUSBANDRY

All standard operating procedures and protocols described in this study plan have been reviewed by Iris Pharma Internal Ethics Committee. All animals were treated according to the Directive 2010/63/UE European Convention for the Protection of Vertebrate Animals used for Experimental [5] and Other Scientific Purposes and to the Association for Research in Vision and Ophthalmology (ARVO) Statement for the Use of Animals in Ophthalmic and Vision Research [6].

7.1. ANIMALS

7.1.1. Animals

Species:	Rat. This is the species most commonly used in this experimental model [1].
Strain:	Lewis (Albino).
Age:	Approximately 8 weeks.
Weight:	130 – 170 g (on ordering), 140 – 170 g (at baseline).
Number/sex:	66 females (study 60; reserve 6).
Breeder:	“JANVIER” – FR-53940 Le Genest Saint Isle.

7.1.2. Identification

Tails were marked using a permanent ink marker, following the inclusion examination.

7.1.3. Health Status

Animals were held in observation for at least 3 days following their arrival. Animals were observed daily for signs of illness and particular attention was paid to the eyes.

7.2. HOUSING

7.2.1. Animal Husbandry

Animals were housed by 2 or 3 in standard cages and under identical environmental conditions. The temperature were held at $22 \pm 2^{\circ}\text{C}$ and the relative humidity at $55 \pm 10\%$. Rooms were continuously ventilated (15 air volumes per hour). Temperature and humidity were continuously controlled and recorded. Animals were routinely exposed (in-cage) to 10-200 lx light in a 12-hour light (from 7:00 a.m. to 7:00 p.m.) and darkness controlled cycle.

7.2.2. Food and Water

Throughout the study, animals had free access to food and water. They were fed a standard dry pellet diet, LASQCDiet® Rod16 (LASVENDI). Tap water was available *ad libitum* from plastic bottles.

8. DESIGN AND PROCEDURE

8.1. STUDY DESIGN

The schedule is presented in **Table 5** hereafter.

Table 5: Study design

Study date	Procedure	Ocular examination	Sampling
Baseline	General clinical examination - Weighing	Slit-lamp	-
D0	General clinical examination Induction of ocular inflammation	-	-
D1 to D6	General clinical examination	-	-
D7	General clinical examination - Weighing Intravenous administration of test and control items Once daily oral administration of reference item	-	-
D8 to D9	General clinical examination Once daily oral administration of reference item	-	-
D10	General clinical examination Intravenous administration of test and control items Once daily oral administration of reference item	-	-
D11	General clinical examination Once daily oral administration of reference item	-	-
D12 to D13	General clinical examination Once daily oral administration of reference item	Slit-lamp for main animals	-
D14	General clinical examination - Weighing Intravenous administration of test and control items Once daily oral administration of reference item		-
D15	General clinical examination Once daily oral administration of reference item		-
D16	General clinical examination		-
D17	General clinical examination	-	-
D18	General clinical examination – Weighing Euthanasia	Slit-lamp for all animals (main + satellites)	- Whole blood (for plasma) - Vitreous (V), choroid/retina (CHR) and iris-ciliary body (ICB)

Treatments and clinical examination were performed in a masked manner.

8.2. EXPERIMENTAL PROCEDURE

8.2.1. Selection of the Animals

Sixty (60) animals were included in this study out of sixty-six (66) ordered. Only animals with no visible sign of ocular defect were randomly assigned to the study groups.

The composition of the groups and the animals' identification are described in **Table 6**.

Table 6: Constitution of the groups

Group No.	Treatment	Dose	Route of administration	Animals' identification	
				Main	Satellites
1	Lpathomab™	30 mg/kg	Twice weekly intravenous administrations (on Days 7, 10 and 14)	1, 2, 3, 4, 5, 6, 7, 8, 9 and 10	11, 12, 13, 14 and 15
2		15 mg/kg		16, 17, 18, 19, 20, 21, 22, 23, 24 and 25	26, 27, 28, 29 and 30
3	Lpathomab™ isotype	30 mg/kg		31, 32, 33, 34, 35, 36, 37, 38, 39 and 40	41, 42, 43, 44 and 45
4	Cyclosporin A	25 mg/kg	Once daily oral administrations from Day 7 to Day 15	46, 47, 48, 49, 50, 51, 52, 53, 54 and 55	56, 57, 58, 59 and 60

*: the vitreous volume in rat is about 0.05 mL

8.2.2. Induction of Ocular Inflammation

On Day 0, ocular inflammation was induced by an injection in the footpad of 100 µL of human S-Ag (100 µg) in Freund's complete adjuvant (2 mg/mL H37Ra (*mycobacterium tuberculosis*)) and an intra-peritoneal injection of 1 µg/100 µL pertussis toxin.

8.2.3. Route and Method of Administration

Test and control items were administered intravenously on Days 7, 10 and 14.

For the reference item, once daily per os administration was performed from Day 7 to Day 15.

8.2.4. Body Weights

Body weights were recorded before the start of in life phase, then on Days 7 and 14 and at the sacrifice.

8.2.5. General In Vivo Observations

The general clinical signs and the appearance of all animals were observed and recorded daily.

8.2.6. Ocular Examination

Both eyes of main animals were examined with a slit-lamp at baseline then once a day from Day 12 to Day 16 and then at the end of the in-life phase, on Day 18, by examiners masked to the treatment groups. Both eyes of satellite animals were examined at baseline and at the end of the in-life phase, on Day 18.

The examination was performed using scales appended in section 14.1.1 page 14. After examination of anterior segment, 10 µL of Mydriaticum® (0.5% tropicamide) were instilled for pupillary dilatation and posterior segment examination.

8.2.7. Euthanasia and Sampling

All the animals were anesthetized by an intramuscular injection of a mixed solution of Rompun® (xylazine) and Imalgene® 1000 (ketamine).

Approximately 5 mL of whole blood (WB) were removed by cardiac puncture and placed into K3-EDTA tubes. After collection, the blood was gently mix by inverting the tube 8 to 10 times, then placed upright on ice until centrifuged at 4°C at 1500 G for 10 minutes. The plasma obtained was transferred into labeled sterile cryogenic vials and stored on ice until transfer to a -80°C freezer.

Animals were euthanized by cardiac injection of overdosed pentobarbital, which is among the recommended methods for euthanasia by the European authorities [5].

Immediately after euthanasia, both eyeballs from all animals were removed and dissected to extract surgically different structures (vitreous, iris-ciliary body and retina-choroid). The ocular tissues of each animals were placed separately in adequately labelled tubes and weighed for further ocular tissue measurements and stored frozen at -80°C.

Plasma samples and ocular tissues were kept frozen at -80°C until shipment (dry ice) to:

LPATH, INC.

c/o Jon Wojciak
4025 Sorrento Valley Blvd
San Diego, CA 92121
USA

9. DATA PROCESSING

Results are expressed in the form of individual and summarized data tables using Microsoft Excel® software.

10. RESULTS

10.1. GENERAL INFORMATION

10.1.1. General Behaviour and Mortality

Between Day 14 (after the third intravenous administration) and Day 17, a high number of animals from groups treated with test and control items were found dead or moribund. For the animals found moribund, it was decided for ethical reasons to sacrifice them and performed sampling as detailed in section 8.2.7 page 10. With the sponsor agreement, the administrations of all study treatments (test, control and reference) and the ocular examinations were discontinued earlier than originally planned. All the rest of the animals (main and satellites) were euthanized on Day 18 after a last ocular examination, and sampled as detailed in section 8.2.7 page 10.

Table 7 below summarizes the day of death with the status of the animals.

Table 7: Day and condition of the death for all the animals

Treatment	Dose	Animal number	Sacrifice day
Lpathomab™	30 mg/kg	3	Found dead on Day 15
		4	Found moribund on Day 16 and then sacrificed
		5	Found dead on Day 17
		6	Found moribund on Day 15 and then sacrificed
		7	Found dead on Day 15
		12	Found dead on Day 16
		15	Found moribund on Day 16 and then sacrificed
		1, 2, 8, 9, 10, 11, 13 and 14	Sacrificed on Day 18
Lpathomab™	15 mg/kg	20	Found dead on Day 16
		24	Found dead on Day 16
		27	Found moribund on Day 17 and then sacrificed
		29	Found dead on Day 16
		16, 17, 18, 19, 21, 22, 23, 25, 26, 28 and 30	Sacrificed on Day 18
Lpathomab™ isotype	30 mg/kg	32	Found dead on Day 16
		33	Found dead on Day 16
		34	Found moribund on Day 15 and then sacrificed
		35	Found dead on Day 16
		36	Found moribund on Day 15 and then sacrificed
		38	Found dead on Day 14
		39	Found dead on Day 16
		41	Found dead on Day 16
		43	Found moribund on Day 15 and then sacrificed
		45	Found moribund on Day 15 and then sacrificed
		31, 37, 40, 42 and 44	Sacrificed on Day 18
Cyclosporin A	25 mg/kg	46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59 and 60	Sacrificed on Day 18

Most of the animals found dead or moribund were treated with Lpathomab™ isotype at 30 mg/kg (10 animals out of 15) and Lpathomab™ at 30 mg/kg (7 animals out of 15).

No abnormal behaviour or other abnormal signs were observed in the treatment arm receiving Cyclosporin A during the study period.

The veterinarian at Iris Pharma performed a necropsy on a moribund animal (rat #43 treated with Lpathomab™ isotype at 30 mg/kg) but no findings were identified that could explain its condition.

For a better understanding, a number of clarifications were asked by the Sponsor. **Table 8** below detailed these questions/clarifications and the answers corresponding.

Table 8: Questions/clarifications

Questions by the Sponsor	Answers by the Study Director
Description of the status of refrigerated condition at arrival of the test articles at Iris.	The test articles were received in good condition, frozen in dry ice. After their receipt, as mentioned in the study plan, the test articles were stored at 2-8°C.
Disposition of test article from Lpath. Dosing of each lot to specific animals.	For animals treated with Lpathomab/LT3114 at 30 mg/kg and 15 mg/kg only the batch number LP121417 was used and for the animals treated with Lpathomab isotype/LT1023, only the batch number LP101210 was used.
Health status of the animals prior to dosing the test article.	Conventional status: SPF animals held in conventional areas.
Handling of the test article at each dosing day after being pulled out of the refrigerator (e.g., allow to reach room temperature).	Allow to reach room temperature.
Water and food consumption before and after dosing of the test articles.	Water or food consumption was not measured during the study. Water and feeding were consumed ad libitum.
Time relationship between dosing with test articles and death.	The study was performed in 2 experimental sets, and the first death was observed the day after the 3rd administration of the 1st set of animals.
Your experience with other antibodies previously injected in Lewis, albino rats in the EAU model, both intravitreally and intravenously.	EAU studies were performed at Iris pharma with antibodies administered intravitreally (both injections) or intravenously. Moreover, another EAU was performed with antibodies administered 4 times (on Days 7, 10, 14 and 17) as this study over the same period of this study with no death being observed.
Necropsy was performed in one animal found dead. Confirm that no finding was reported that could provide any indication of likely cause of death.	No finding that could provide any indication of likely cause of death.
Confirm lack of swelling and redness in paw, ears and other parts that could indicate an anaphylactic reaction had occurred.	No swelling or redness in ears and other parts. However, since the induction was performed in the footpad, all the animals showed a noticeable swelling for this foot, as expected.
Confirm that all affected animals started to display distress only after the third injection with the test articles but not immediately following IV dosing.	Confirmed. No dead or moribund animals were observed after the first and the second injections.
Confirm that no obvious precipitates or other unusual findings in the test articles were noticed.	All solutions were clear.

In conclusion, no reason was identified that could explain the mortality rates observed in the two test article arms and in the isotype control arm.

10.1.2. Body Weights

Individual body weights are presented in section **14.2** page **15**.

The mean body weights remained stable during the study, except for the group treated with Cyclosporin A that showed a body weight gain within usual values.

10.2. OCULAR EVALUATION

Individual data, summarized in section 14.3.1 page 17 are reported in section 14.3.2 pages 18 to 21.

As a result of the high mortality rate and early discontinuation of study treatments, no conclusions could be drawn regarding the potential effect of Lpathomab™ on intraocular inflammation regardless of the dose.

11. GENERAL CONCLUSION

A high mortality was observed between Day 14 and Day 17 in groups treated with Lpathomab™ isotype at 30 mg/kg (10 animals found dead or moribund out of 15) and Lpathomab™ at 30 mg/kg (7 animals found dead or moribund out of 15). For the group treated with Lpathomab™ at 15 mg/kg, the mortality was moderate (4 animals found dead or moribund out of 15). No reason was found to explain the observed mortality.

As a result of the high mortality rate and early discontinuation of study treatments, no conclusions could be drawn regarding the potential effect of Lpathomab™ on intraocular inflammation regardless of the dose.

12. ARCHIVING OF DATA

All source documents, raw data, final study plan and report will be archived at IRIS PHARMA for 5 years following the date of the final report. At the end of the 5-year period, IRIS PHARMA will contact the Sponsor for instructions on the transfer, retention or disposal of these documents. Fees for the transfer or continued retention of these materials following the 5-year archive period will be invoiced to the Sponsor.

13. REFERENCES

1. CASPI RR (2006) Mechanisms underlying autoimmune uveitis. *Drug Discov Today: Disease Mechanism* 3:199-206.
2. NUSSENBLATT RB (1991) Proctor lecture. Experimental autoimmune uveitis: mechanisms of disease and clinical therapeutic indications. *Invest Ophthalmol Vis Sci* 32(13): 3131-3141.
3. NUSSENBLATT RB, M. M. R., WACKER W.B., CEVARIO S.J, SALINAS-CARMONA M.C, and GERY I. (1981). "Cyclosporin A. Inhibition of experimental autoimmune uveitis in Lewis rats." *J Clin Invest.* 67(4): 1228-1231.
4. MOCHIZUKI M, N. R., KUWABARA T, GERY I. (1985). "Effects of cyclosporine and other immunosuppressive drugs on experimental autoimmune uveoretinitis in rats." *Invest Ophthalmol Vis Sci.* 26(2): 226-232.
5. FRENCH DECREE N° 2001-486, DATED JUNE 06, 2001 PUBLISHING THE EUROPEAN CONVENTION ON ANIMAL PROTECTION FOR SCIENTIFIC EXPERIMENTATION OF 1986. *J. Offic. Rep. Fr.* 2001; 133: 9094-9120 (*Offic. J. Eur. Commun.* 1999; 42: L222, 31-37).
6. ASSOCIATION FOR RESEARCH IN VISION AND OPHTHALMOLOGY (ARVO). Statement for the Use of Animals in Ophthalmic and Vision Research.

14. APPENDICES

14.1. EVALUATION'S SCALES

Ref. CAMELO and al, JOPT Vol 25 p 9-16.

14.1.1. Scale for Ocular Examination

- No sign of inflammation, normal iris dilatation after instillation with a mydriatic drug.....0
- Discrete inflammation in iris and conjunctiva.....1
- Dilatation of iris and conjunctival vessels.....2
- Hyperhemia in iris associated with the Tyndall effect in anterior chamber.....3
- + 1 point was added if synechia, fibrin, or hypopion (cell deposit in the inferior anterior chamber) or cells in anterior chamber were observed

14.1.2. Fundus Examination

- Normal retina.....0
- Exudative retinal edema and/or hemorrhagia.....1
- ND: inability to observe the fundus due to an anterior intraocular inflammation involvement.

14.2. BODY WEIGHTS (INDIVIDUAL DATA)

Table 9: Group treated with Lpathomab™ at 30 mg/kg

	Rat n°	Body weight (g)			
		Time-point			
		Baseline	Day 7	Day 14	Day of sacrifice (Day 15 - Day 18)
Lpathomab™ 30 mg/kg intravenous injection	1	158	151	162	156
	2	143	142	158	148
	3	154	168	166	†
	4	158	164	164	161
	5	156	154	154	†
	6	148	163	154	†
	7	147	166	169	†
	8	158	163	164	160
	9	156	155	155	164
	10	156	167	168	163
	11	160	161	166	164
	12	143	170	168	†
	13	142	153	153	160
	14	155	162	157	153
	15	165	176	167	165
	Mean	153.3	161.0	161.7	159.4
	SD	7.0	8.6	5.9	5.5

SD = standard deviation

† = animal dead

Table 10: Group treated with Lpathomab™ at 15 mg/kg

	Rat n°	Body weight (g)			
		Time-point			
		Baseline	Day 7	Day 14	Day of sacrifice (Day 16 - Day 18)
Lpathomab™ 15 mg/kg intravenous injection	16	150	161	158	158
	17	151	162	166	164
	18	152	161	164	160
	19	160	169	164	160
	20	152	162	158	†
	21	168	165	163	160
	22	154	165	162	159
	23	162	168	164	165
	24	150	170	167	†
	25	158	168	167	164
	26	153	163	168	166
	27	148	159	162	163
	28	147	160	170	162
	29	148	166	164	†
	30	147	169	162	167
	Mean	153.3	164.5	163.9	162.3
	SD	6.1	3.7	3.4	2.9

SD = standard deviation

† = animal dead

Table 11: Group treated with Lpathomab™ isotype at 30 mg/kg

	Rat n°	Body weight (g)			
		Time-point			
		Baseline	Day 7	Day 14	Day of sacrifice (Day 14 - Day 18)
Lpathomab™ isotype 30 mg/kg intravenous injection	31	145	161	166	166
	32	148	156	158	†
	33	150	167	166	†
	34	160	168	166	†
	35	149	161	162	†
	36	147	158	164	155
	37	148	159	165	169
	38	154	169	165	†
	39	142	163	159	†
	40	141	150	162	166
	41	151	170	174	†
	42	156	165	170	170
	43	157	166	166	†
	44	142	142	149	164
	45	167	173	170	169
	Mean	150.5	161.9	164.1	165.6
	SD	7.3	8.1	5.9	5.1

SD = standard deviation

† = animal dead

Table 12: Group treated with Cyclosporin A

	Rat n°	Body weight (g)			
		Time-point			
		Baseline	Day 7	Day 14	Day of sacrifice (Day 18)
Cyclosporin A 25 mg/mL	46	148	162	170	180
	47	149	148	162	170
	48	149	166	150	170
	49	147	155	148	168
	50	162	164	156	176
	51	162	172	169	181
	52	149	163	168	178
	53	156	172	179	187
	54	153	161	166	180
	55	155	159	170	171
	56	148	164	174	178
	57	160	169	176	176
	58	150	154	166	170
	59	152	162	169	176
	60	160	169	180	181
	Mean	153.3	162.7	166.9	176.1
	SD	5.4	6.7	9.5	5.4

SD = standard deviation

14.3. SLIT-LAMP OCULAR EVALUATION

14.3.1. Summarized data

Table 13: Main animals

		Ocular evaluation both eyes (scale 0 - 7)						
Treatment		Baseline	Day 12	Day 13	Day 14	Day 15	Day 16	Day 18
Lpathomab™ 30 mg/kg intravenous injection	Mean	0.0	0.0	0.0	0.3	0.8	1.3	1.5
	SD	0.0	0.0	0.0	1.1	2.0	2.3	2.2
	Incidence %	0.0	0.0	0.0	5.0	14.3	25.0	40.0
Lpathomab™ 15 mg/kg intravenous injection	Mean	0.0	0.1	0.5	0.4	0.5	0.5	0.9
	SD	0.0	0.4	1.4	1.2	1.1	1.5	1.8
	Incidence %	0.0	5.0	20.0	15.0	20.0	18.8	25.0
Lpathomab™ isotype 30 mg/kg intravenous injection	Mean	0.0	0.3	0.3	0.5	0.4	0.2	0.0
	SD	0.0	1.3	1.3	1.3	1.3	0.4	0.0
	Incidence %	0.0	5.0	5.0	15.0	11.1	16.7	0.0
Cyclosporine A 25 mg/mL	Mean	0.0	0.0	0.0	0.1	0.1	0.1	0.3
	SD	0.0	0.0	0.0	0.2	0.3	0.2	0.8
	Incidence %	0.0	0.0	0.0	5.0	10.0	5.0	10.0

SD = standard deviation

Incidence = proportion of eyes with a score greater than 0

Table 14: Satellite animals

		Ocular evaluation both eyes (scale 0 - 7)	
Treatment		Baseline	Day 18
Lpathomab™ 30 mg/kg intravenous injection	Mean	0.0	0.0
	SD	0.0	0.0
	Incidence %	0.0	0.0
Lpathomab™ 15 mg/kg intravenous injection	Mean	0.0	0.2
	SD	0.0	0.4
	Incidence %	0.0	16.7
Lpathomab™ isotype 30 mg/kg intravenous injection	Mean	0.0	0.0
	SD	0.0	0.0
	Incidence %	0.0	0.0
Cyclosporin A 25 mg/mL	Mean	0.0	0.1
	SD	0.0	0.3
	Incidence %	0.0	10.0

SD = standard deviation

Incidence = proportion of eyes with a score greater than 0

14.3.2. Individual data

14.3.2.1. Ocular evaluation

Table 15: Group treated with Lpathomab™ at 30 mg/kg (main animals)

	Rat n°	Ocular evaluation (scale 0 - 7)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
		Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Lpathomab™ 30 mg/kg intravenous injection	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0
	3	0	0	0	0	0	0	0	0	†		†		†	
	4	0	0	0	0	0	0	0	0	0	0	†		†	
	5	0	0	0	0	0	0	0	0	0	0	0	0	†	
	6	0	0	0	0	0	0	0	0	†		†		†	
	7	0	0	0	0	0	0	0	0	†		†		†	
	8	0	0	0	0	0	0	0	0	0	6	0	5	0	4
	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	10	0	0	0	0	0	0	0	5	0	5	5	5	5	5
	Mean	0.0		0.0		0.0		0.3		0.8		1.3		1.5	
	SD	0.0		0.0		0.0		1.1		2.0		2.3		2.2	
	Incidence %	0.0		0.0		0.0		5.0		14.3		25.0		40.0	

SD = standard deviation †= animal dead Incidence = proportion of eyes with a score greater than 0

Table 16: Group treated with Lpathomab™ at 30 mg/kg (satellite animals)

	Rat n°	Ocular evaluation (scale 0 - 7)			
		Baseline		Day 18	
		Right eye	Left eye	Right eye	Left eye
Lpathomab™ 30 mg/kg intravenous injection	11	0	0	0	0
	12	0	0	†	
	13	0	0	0	0
	14	0	0	0	0
	15	0	0	†	
	Mean	0.0		0.0	
	SD	0.0		0.0	
	Incidence %	0.0		0.0	

SD = standard deviation †= animal dead Incidence = proportion of eyes with a score greater than 0

Table 17: Group treated Lpathomab™ at 15 mg/kg (main animals)

	Rat n°	Ocular evaluation (scale 0 - 7)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
		Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Lpathomab™ 15 mg/kg intravenous injection	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	17	0	0	0	0	0	6	0	5	0	4	0	6	4	4
	18	0	0	0	0	0	0	0	0	0	0	0	0	5	0
	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	20	0	0	2	0	2	0	2	0	2	0	†		†	
	21	0	0	0	0	1	0	0	0	2	0	1	0	2	0
	22	0	0	0	0	1	0	1	0	0	0	0	1	0	0
	23	0	0	0	0	0	0	0	0	0	1	0	0	0	0
	24	0	0	0	0	0	0	0	0	0	0	†		†	
	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mean	0.0		0.1		0.5		0.4		0.5		0.5		0.9	
	SD	0.0		0.4		1.4		1.2		1.1		1.5		1.8	
	Incidence %	0.0		5.0		20.0		15.0		20.0		18.8		25.0	

SD = standard deviation † = animal dead Incidence = proportion of eyes with a score greater than 0

Table 18: Group treated Lpathomab™ at 15 mg/kg (satellite animals)

	Rat n°	Baseline		Day 18	
		Right eye	Left eye	Right eye	Left eye
Lpathomab™ 15 mg/kg intravenous injection	26	0	0	0	0
	27	0	0	†	
	28	0	0	0	1
	29	0	0	†	
	30	0	0	0	0
	Mean	0.0		0.2	
	SD	0.0		0.4	
	Incidence %	0.0		16.7	

SD = standard deviation † = animal dead Incidence = proportion of eyes with a score greater than 0

Table 19: Group treated with Lpathomab™ isotype at 30 mg/kg (main animals)

		Ocular evaluation (scale 0 - 7)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
	Rat n°	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Lpathomab™ isotype 30 mg/kg intravenous injection	31	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	32	0	0	0	0	0	0	0	0	0	0	†		†	
	33	0	0	0	0	0	0	0	0	0	0	†		†	
	34	0	0	0	0	0	0	3	0	3	0	†		†	
	35	0	0	0	0	0	0	0	0	0	0	†		†	
	36	0	0	6	0	6	0	5	1	5	0	†		†	
	37	0	0	0	0	0	0	0	0	0	0	1	0	0	0
	38	0	0	0	0	0	0	0	0	†		†		†	
	39	0	0	0	0	0	0	0	0	0	0	†		†	
	40	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mean	0.0		0.3		0.3		0.5		0.4		0.2		0.0	
	SD	0.0		1.3		1.3		1.3		1.3		0.4		0.0	
	Incidence %	0.0		5.0		5.0		15.0		11.1		16.7		0.0	

SD = standard deviation †= animal dead Incidence = proportion of eyes with a score greater than 0

Table 20: Group treated with Lpathomab™ isotype at 30 mg/kg (satellite animals)

		Baseline		Day 18	
	Rat n°	Right eye	Left eye	Right eye	Left eye
Lpathomab™ isotype 30 mg/kg intravenous injection	41	0	0	†	
	42	0	0	0	0
	43	0	0	†	
	44	0	0	0	0
	45	0	0	†	
	Mean	0.0		0.0	
	SD	0.0		0.0	
	Incidence %	0.0		0.0	

SD = standard deviation †= animal dead Incidence = proportion of eyes with a score greater than 0

Table 21: Group treated with Cyclosporin A (main animals)

		Ocular evaluation (scale 0 - 7)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
	Rat n°	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Cyclosporin A 25 mg/mL	46	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	47	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	48	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	49	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	50	0	0	0	0	0	0	0	0	1	1	0	0	0	0
	51	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	52	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	53	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	54	0	0	0	0	0	0	0	1	0	0	0	1	2	3
	55	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mean	0.0		0.0		0.0		0.1		0.1		0.1		0.3	
	SD	0.0		0.0		0.0		0.2		0.3		0.2		0.8	
	Incidence %	0.0		0.0		0.0		5.0		10.0		5.0		10.0	

SD = standard deviation

Incidence = proportion of eyes with a score greater than 0

Table 22: Group treated with Cyclosporin A (satellite animals)

		Ocular evaluation (scale 0 - 7)			
		Baseline		Day 18	
	Rat n°	Right eye	Left eye	Right eye	Left eye
Cyclosporin A 25 mg/mL	56	0	0	0	0
	57	0	0	0	0
	58	0	0	0	0
	59	0	0	1	0
	60	0	0	0	0
	Mean	0.0		0.1	
	SD	0.0		0.3	
	Incidence %	0.0		10.0	

SD = standard deviation Incidence = proportion of eyes with a score greater than 0

14.3.2.2. Fundus examination

Table 23: Group treated with Lpathomab™ at 30 mg/kg (main animals)

		Fundus (scale 0 - 1)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
	Rat n°	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Lpathomab™ 30 mg/kg intravenous injection	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	3	0	0	0	0	0	0	0	0		†		†		†
	4	0	0	0	0	0	0	0	0	0	0		†		†
	5	0	0	0	0	0	0	0	0	0	0	0	0		†
	6	0	0	0	0	0	0	0	0		†		†		†
	7	0	0	0	0	0	0	0	0		†		†		†
	8	0	0	0	0	0	0	0	0	0	0	0	1	0	1
	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	10	0	0	0	0	0	1	0	1	0	1	1	1	1	1
	Mean	0.0		0.0		0.1		0.1		0.1		0.3		0.3	
	SD	0.0		0.0		0.2		0.2		0.3		0.5		0.5	
	Incidence %	0.0		0.0		5.0		5.0		7.1		25.0		30.0	

SD = standard deviation † = animal dead Incidence = proportion of eyes with a score greater than 0

Table 24: Group treated with Lpathomab™ at 30 mg/kg (satellite animals)

		Fundus (scale 0 - 1)			
		Baseline		Day 18	
	Rat n°	Right eye	Left eye	Right eye	Left eye
Lpathomab™ 30 mg/kg intravenous injection	11	0	0	0	0
	12	0	0		†
	13	0	0	0	0
	14	0	0	0	0
	15	0	0		†
	Mean	0.0		0.0	
	SD	0.0		0.0	
	Incidence %	0.0		0.0	

SD = standard deviation † = animal dead Incidence = proportion of eyes with a score greater than 0

Table 25: Group treated with Lpathomab™ at 15 mg/kg (main animals)

	Rat n°	Fundus (scale 0 - 1)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
		Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Lpathomab™ 15 mg/kg intravenous injection	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	17	0	0	0	0	0	1	0	1	0	1	0	0	0	0
	18	0	0	0	0	0	0	0	0	0	0	0	0	1	0
	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	20	0	0	0	0	0	0	0	0	0	0	†		†	
	21	0	0	0	0	1	0	0	0	0	0	0	0	0	0
	22	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	23	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	24	0	0	0	0	0	0	0	0	0	0	†		†	
	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mean	0.0		0.0		0.1		0.1		0.1		0.0		0.1	
	SD	0.0		0.0		0.3		0.2		0.2		0.0		0.3	
	Incidence %	0.0		0.0		10.0		5.0		5.0		0.0		6.3	

SD = standard deviation †= animal dead Incidence = proportion of eyes with a score greater than 0

Table 26: Group treated with Lpathomab™ at 15 mg/kg (satellite animals)

	Rat n°	Fundus (scale 0 - 1)			
		Baseline		Day 18	
		Right eye	Left eye	Right eye	Left eye
Lpathomab™ 15 mg/kg intravenous injection	26	0	0	0	0
	27	0	0	†	
	28	0	0	0	0
	29	0	0	†	
	30	0	0	0	0
	Mean	0.0		0.0	
	SD	0.0		0.0	
	Incidence %	0.0		0.0	

SD = standard deviation †= animal dead Incidence = proportion of eyes with a score greater than 0

Table 27: Group treated with Lpathomab™ isotype at 30 mg/kg (main animals)

	Rat n°	Fundus (scale 0 - 1)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
		Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Lpathomab™ isotype 30 mg/kg intravenous injection	31	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	32	0	0	0	0	0	0	0	0	0	0	†		†	
	33	0	0	0	0	0	0	0	0	0	0	†		†	
	34	0	0	0	0	0	0	1	0	0	0	†		†	
	35	0	0	0	0	0	0	0	0	0	0	†		†	
	36	0	0	1	0	ND	0	1	0	1	0	†		†	
	37	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	38	0	0	0	0	0	0	0	0	†		†		†	
	39	0	0	0	0	0	0	0	0	0	0	†		†	
	40	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mean	0.0		0.1		0.0		0.1		0.1		0.0		0.0	
	SD	0.0		0.2		0.0		0.3		0.2		0.0		0.0	
	Incidence %	0.0		5.0		0.0		10.0		5.6		0.0		0.0	

SD = standard deviation ND = not determined † = animal dead Incidence = proportion of eyes with a score greater than 0

Table 28: Group treated with Lpathomab™ isotype at 30 mg/kg (satellite animals)

	Rat n°	Fundus (scale 0 - 1)			
		Baseline		Day 18	
		Right eye	Left eye	Right eye	Left eye
Lpathomab™ isotype 30 mg/kg intravenous injection	41	0	0	†	
	42	0	0	0	0
	43	0	0	†	
	44	0	0	0	0
	45	0	0	†	
	Mean	0.0		0.0	
	SD	0.0		0.0	
	Incidence %	0.0		0.0	

SD = standard deviation † = animal dead Incidence = proportion of eyes with a score greater than 0

Table 29: Group treated with Cyclosporin A (main animals)

	Rat n°	Fundus (scale 0 - 1)													
		Baseline		Day 12		Day 13		Day 14		Day 15		Day 16		Day 18	
		Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye	Right eye	Left eye
Cyclosporin A 25 mg/mL	46	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	47	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	48	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	49	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	51	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	52	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	53	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	54	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	55	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mean	0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	SD	0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Incidence %	0.0		0.0		0.0		0.0		0.0		0.0		0.0	

SD = standard deviation

Incidence = proportion of eyes with a score greater than 0

Table 30: Group treated with Cyclosporin A (satellite animals)

	Rat n°	Fundus (scale 0 - 1)			
		Baseline		Day 18	
		Right eye	Left eye	Right eye	Left eye
Cyclosporin A 25 mg/mL	56	0	0	0	0
	57	0	0	0	0
	58	0	0	0	0
	59	0	0	0	0
	60	0	0	0	0
	Mean	0.0		0.0	
	SD	0.0		0.0	
	Incidence %	0.0		0.0	

SD = standard deviation

Incidence = proportion of eyes with a score greater than 0

14.3.2.3. Ocular observations (comments)

Only animals with observations are presented

Table 31: Group treated with Lpathomab™ at 30 mg/kg (main animals)

	Rat n°	Right eye			Left eye					
		Baseline	Day 16	Day 18	Baseline	Day 13	Day 14	Day 15	Day 16	Day 18
Lpathomab™ 30 mg/kg intravenous injection	2	-	-	Fibrin	-	-	-	-	-	-
	5	-	-	†	-	-	-	Lens default	Lens default	†
	8	-	-	-	-	-	-	Synechia + Fibrin + Myosis	Fibrin + Myosis + Retinal detachment	Fibrin + Retinal detachment
	10	-	Fibrin + Myosis + Retinal detachment	Fibrin + Myosis + Retinal detachment	-	Retinal detachment	Fibrin + Myosis + Retinal detachment	Fibrin + Myosis + Retinal detachment + yellow eye	Fibrin + Myosis + Retinal detachment	Fibrin + Myosis + Retinal detachment

"-" = nothing to report

† = animal dead

Table 32: Group treated with Lpathomab™ at 15 mg/kg (main animals)

	Rat n°	Right eye						
		Baseline	Day 12	Day 13	Day 14	Day 15	Day 16	Day 18
Lpathomab™ 15 mg/kg intravenous injection	17	-	-	-	-	-	-	Fibrin
	18	-	-	-	-	-	-	Fibrin + Myosis + Retinal detachment
	20	-	Synechia + Fibrin	Synechia + Fibrin	Synechia + Fibrin	Synechia + Fibrin + Lens default	†	†
	21	-	-	Partial Mydriasis + Retinal detachment	-	Synechia + Fibrin	Fibrin	Fibrin
	22	-	-	Fibrin	Fibrin	-	-	-
	23	-	-	-	-	-	-	-

	Rat n°	Left eye					
		Baseline	Day 13	Day 14	Day 15	Day 16	Day 18
Lpathomab™ 15 mg/kg intravenous injection	17	-	Synechia + Fibrin + Myosis + Retinal detachment	Fibrin + Myosis + Retinal detachment	Fibrin + Retinal detachment	Synechia + Fibrin + Myosis	Fibrin
	18	-	-	-	-	-	-
	20	-	-	-	-	†	†
	21	-	-	-	-	-	-
	22	-	-	-	-	Fibrin	-
	23	-	-	-	Fibrin	-	-

"-" = nothing to report

† = animal dead

Table 33: Group treated with Lpathomab™ at 15 mg/kg (satellite animals)

	Rat n°	Right eye		Left eye	
		Baseline	Day 18	Baseline	Day 18
Lpathomab™ 15 mg/kg intravenous injection	28	-	-	-	Fibrin

"-" = nothing to report

Table 34: Group treated with Lpathomab™ isotype at 30 mg/kg (main animals)

	Rat n°	Right eye							Left eye				
		Baseline	Day 12	Day 13	Day 14	Day 15	Day 16	Day 18	Baseline	Day 14	Day 15	Day 16	Day 18
Lpathomab™ isotype 30 mg/kg intravenous injection	34	-	-	-	Synechia + Fibrin + Partial mydriasis + Retinal detachment	Fibrin + Yellow eye	†	†	-	-	-	†	†
	36	-	Synechia + Fibrin + Cellular deposit + Retinal detachment	Synechia + Fibrin + Myosis	Fibrin + Myosis + Retinal detachment	Fibrin + Myosis + Retinal detachment	†	†	-	Cell deposit	Yellow eye	†	†
	37	-	-	-	-	-	Fibrin	-	-	-	-	-	-

"-" = nothing to report

† = animal dead

Table 35: Group treated with Cyclosporin A (main animals)

	Rat n°	Left eye		
		Baseline	Day 14	Day 16
Cyclosporin A 25 mg/mL	54	-	Fibrin	Fibrin

"-" = nothing to report

Table 36: Group treated with Cyclosporin A (satellite animals)

	Rat n°	Right eye	
		Baseline	Day 18
Cyclosporin A 25 mg/mL	59	-	Synechia

"-" = nothing to report